Exact Compositional Transfer of Bounded Linear Recovery

Tavis Rudd

August 2026

Abstract

Bounded linear recovery under concatenation is not determined by scalar helper thresholds: an outer code can force intermediate functionals that scalar minimization forgets. Shortening and puncturing the inner dual onto the helpers give a canonical pair $K_P \subseteq D_P$. Its relative generalized Hamming weights are exact minimum helper unions but omit the labels needed for finite transfer. We prove that minimizing labelled prescribed-coset support costs over the complete outer functional dual gives $\Gamma_{j,t}(O, I)$, the minimum cost of a rank-t nonconfined recovery system, without a distance hypothesis. These functions compose by min–sum through finite towers; stored argmins propagate an optimizing coefficient witness. Rank-one probes characterize exactly which numerical distinctions arbitrary outer contexts can observe, while rank-one confinement controls simultaneous confinement at every recoverable rank. When outer distance removes the nonzero sectors, the exact cost collapses to $M_t(D_P, K_P) + d(I^\perp)$. Below a confinement threshold, restriction and zero-extension preserve every normalized equation and exact support. Equal relative-weight hierarchies need not determine bounded reliability. A companion exact compiler executes the witness-returning recursion; proofs are software-independent.

1 Introduction

Which recovery information survives concatenation? A scalar locality or confinement threshold does not: the outer code can force an intermediate functional whose cost was discarded when the scalar minimum was taken. The problem is therefore compositional. We seek finite data that determine exactly when a recovery equation can leave its inner block and that can be propagated through another concatenation level.

A linear recovery equation expresses a target combination as a linear combination of helper symbols; its cost is the number of distinct helpers used.

Here exact transfer means more than equality of a numerical parameter. Below a helper radius, restriction to one inner block and zero-extension back into the concatenation must be inverse bijections on normalized recovery-equation systems, preserving their coefficients and exact helper supports. This is the level of structure needed to transport recovery-set overlap, bounded reliability, and capacity constraints.

Three projections of the recovery data answer different questions. Relative generalized Hamming weights give the minimum helper union at each recovered dimension. Exact helper-support families retain the overlap data needed for reliability. Labelled prescribed-coset support functions keep the intermediate functional labels needed for concatenation. Figure 1 shows how these projections meet; neither the support-family branch nor the labelled-cost branch contains the other.

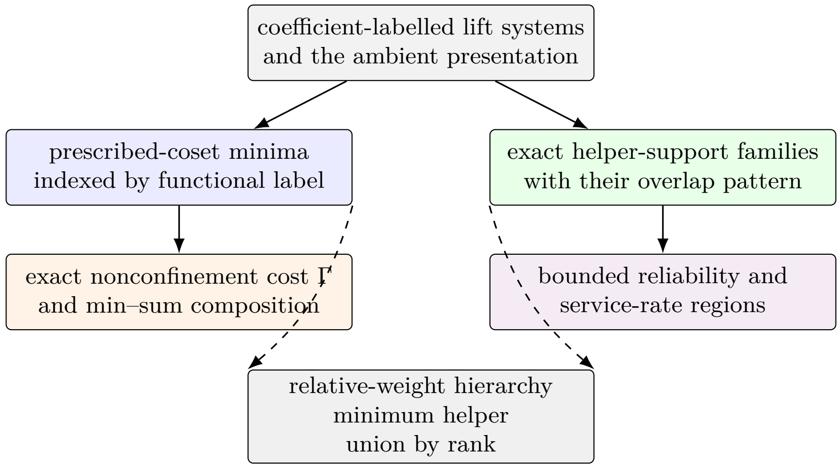

Figure 1: The two information branches used in the paper. Fibrewise labelled minima form an exact state for numerical composition; minimizing lifts must be stored separately to propagate coefficient witnesses. Exact support families retain the overlaps needed for stochastic and fractional recovery questions. Relative weights are the common minimum-cost projection (dashed arrows), complete for helper-union minima but not for either richer question.

Let q be a prime power, let $I \leq \mathbb{F}_q^E$, and partition the coordinates as $E = P \sqcup J$ into targets and helpers. The associated nested code pair is

$$D_P = \mathrm{punct}_J(I^\perp), \qquad K_P = \mathrm{short}_J(I^\perp).$$

For a full-row-rank encoder $G = (G_P \mid G_J)$, put $W_P = \operatorname{im} G_P \cap \operatorname{im} G_J$. The exact sequence in Proposition 3.1 identifies $D_P/K_P \cong W_P$, the space of target combinations recoverable from the helpers. We assume $0 < \dim I < |E|$, so both the represented message space and $I^\perp$ are nonzero. We use the convention $d(0) = \infty$ for the zero code. Here t is the dimension of a recovered subspace of W_P, not the number of erased coordinate positions. Recovering all values on P is the special case $\operatorname{im} G_P \subseteq \operatorname{im} G_J$ and $t = \dim \operatorname{im} G_P$; dependent target columns may make this dimension smaller than $|P|$.

The first layer is classical nested-code support theory in recovery language: the minimum helper union for recovered dimension t is $M_t(D_P, K_P)$. Theorem 3.3 gives the exact support-preserving correspondence. The principal result begins at the next layer, where the outer code can distinguish functional labels having the same minimum support.

For an outer code O, a fixed target block j, and a nonzero recovered subspace $T \leq W_P$, Section 4 defines $\Gamma_{j,T}(O, I)$ by minimizing the target-normalized cost in block j and the ordinary prescribed-coset costs in the other blocks over the complete outer functional dual. Put $\Gamma_{j,t}(O, I) = \min\{\Gamma_{j,T}(O, I) : T \leq W_P,\ \dim T = t\}$.

Theorem 1.1 (Main characterization: exact finite transfer)**.** *Let $\ell = \dim W_P \geq 1$ and $1 \leq t \leq \ell$. Let I be used as the represented inner code in a concatenation with an L-linear outer code $O \leq L^N$ with $N \geq 2$, where L is the message space of I, and fix a block j onto which O projects nontrivially. The minimum cost of a nonconfined recovery system in recovered dimension t is $\Gamma_{j,t}(O, I)$. Hence confinement through radius r holds exactly when $r < \Gamma_{j,t}(O, I)$.*

Outer-distance collapse. The exact characterization gives

$$d(O^\perp) \geq M_t(D_P, K_P) + d(I^\perp) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^\perp).$$

For a fixed radius r, the weaker condition $d(O^{\perp}) > r+1$ reduces the exact criterion to

$$r < M_t(D_P, K_P) + d(I^{\perp}).$$

When this outer-distance condition and the displayed inequality both hold, zero-extension gives a bijection on the normalized recovery-equation systems and their exact helper supports. Consequently, for any outer family with dual distance tending to infinity, the same statement holds for all sufficiently long concatenations at every fixed r.

The second central result is the closure law in Theorem 4.3: the labelled prescribed-coset functions entering $\Gamma_{j,T}$ compose associatively by exact min–sum substitution. The numerical functions retain the minimum in each labelled fibre. If the recursion also stores a minimizing lift in each attained fibre, those argmins propagate an optimizing coefficient-level witness through the tower. Transporting the full family instead requires the full lift sets used in the proof of the closure theorem.

Three exact consequences sharpen this state description. Projective line probes characterize the rank-one numerical contextual quotient, without claiming that the raw labelled table itself is minimal (Theorem 4.4). At target rank t, outer functional duals of dimension at most t test every outer context (Proposition 4.5). Finally, rank-one confinement through a fixed radius is equivalent to confinement at every recoverable rank, with the complete bounded coefficient and support families transported simultaneously (Corollary 4.6).

The mechanism has two exhaustive sectors. A zero outer-functional label gives one inner recovery system plus a minimum inner-dual perturbation in another block. A nonzero label forces prescribed cosets in the blocks selected by the outer functional dual. The minimum over the zero and nonzero sectors is $\Gamma_{j,T}(O, I)$. Outer dual distance does not create the additive RGHW formula; it removes the nonzero sector and leaves that formula as the surviving zero-sector cost.

The label loss is already visible over $\mathbb{F}_2 \subseteq \mathbb{F}_4$. Two three-column encoders have the same binary image code, dual, abstract nested pair, complete relative-weight hierarchy, dual distance, and unlabelled helper-support family. Their functional labels differ, and one fixed outer functional-dual line gives exact nonconfinement costs one and two. Proposition 4.2 gives the calculation before the closure theorem. Thus every unlabelled and scalar explanation is held fixed; only alignment with the intermediate functional labels changes.

The restriction–extension bijection yields the later transfer results. It preserves every block-local statistic determined by normalized recovery systems of cost at most r. After forgetting coefficients, it also preserves statistics determined by inclusion-minimal helper supports, including bounded repair reliability and the bounded service-rate region. This statement is block-local: the upward-closed recovery-set family in the full concatenated coordinate set also contains irrelevant cross-block supersets.

The theorem contains the previous one-coordinate criterion. If $P = \{x\}$ and $t = 1$, then $M_1(D_P, K_P)$ is the least number of helpers in a dual word through x, and

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^{\perp}).$$

Thus $r < M_1 + d(I^{\perp})$ is exactly $r + 1 < z_x(I)$.

Relative generalized Hamming weights (RGHWs) and relative dimension/length profiles (RDLPs) already describe support and access thresholds for nested codes in wiretap and ramp-secret-sharing settings [19, 15, 11, 28, 10]. The identity $\mu_t(P) = M_t(D_P, K_P)$ is this standard invariant specialized to the canonical target/helper pair. Recovery-set and locality formulations retain helper supports but generally discard the normalized coefficients and functional labels that couple concatenation blocks [27, 12, 25, 1, 21]. Work on recovering entire subspaces from simplex-code columns optimizes disjoint support families, not their coefficient-labelled transfer through concatenation [5]. Regenerating-code models optimize download and subpacketization rather than the scalar helper-support cost used here [29, 31, 16].

Concatenation has long been used to construct locally repairable codes (LRCs) with prescribed parameter properties [17, 14]. Jin and Fu fix a binary $[3, 2, 2]$ inner code and choose $\mathbb{F}_4$ outer codes to obtain parameter-optimal binary locality-two codes. Here the inner code and target split are arbitrary, and the question is the exact obstruction to preserving every bounded dual recovery equation. The finite formula is rank-stratified and ungated; the RGHW formula is its outer-distance specialization. We make no minimum-bandwidth claim. The functional-dual decomposition is classical concatenation machinery, and min–sum elimination also appears in soft and generalized-concatenated decoding [9, 6, 2, 13, 3]. Label-refined precedents are also important: soft concatenated decoding passes symbol-indexed reliabilities, and input–output weight enumerators can depend on the chosen generator presentation and compose by convolution in recursive constructions [13, 18]. Nogin's message weight functions provide a related representation-sensitive refinement of ordinary weight data [24]. These precedents retain labels for decoding or enumeration; they do not impose target normalization or identify the exact first nonconfined support. Here target-normalized prescribed-coset costs characterize that support and form a closed state under repeated concatenation.

The terminology "contextual equivalence" and "congruence" describes the response to all permitted outer compositions. It parallels behavioural minimization for weighted automata over semirings [23], but we make no new general automata-minimization claim: Theorem 4.4 identifies the separating outer-code probes for this finite coding problem.

ergodis (Exact Recovery, Global Optimization, and Invariant Synthesis) turns these reductions into an exact compiler and solver. It composes labelled costs through finite towers, returns recovery witnesses and confinement thresholds, and optimizes simultaneous repairs under heterogeneous capacities. Its bundled application examples also cover recursive XOR repair, LRC batching, repair DAGs, QC-LDPC search, vector repair, and MDS checkpoint recovery. No mathematical result depends on the implementation or its measurements. Section 6 gives correctness and complexity bounds and compares matched exact controls for the storage, graph, coding, and tower examples; CP-SAT is one control and a residual backend, not the sole comparison target.

2 Recovery sets and normalized equations

Let $C \leq \mathbb{F}_q^E$ be a linear code. We first separate four layers that are often harmlessly conflated when only locality is at issue.

Fix a target $x \in E$ and a helper bound r. A dual word $w \in C^\perp$ with $w_x \neq 0$ gives

$$c_x = \sum_{y \neq x} -\frac{w_y}{w_x} c_y \qquad (c \in C).$$

Its exact helper support is $\operatorname{supp}(w) \setminus \{x\}$. Define

$$\mathcal{H}_x^{(\leq r)}(C) = \{\operatorname{supp}(w) \setminus \{x\} : w \in C^\perp, w_x \neq 0, \operatorname{wt}(w) \leq r + 1\}.$$

The usual bounded recovery-set family is the upward closure of $\mathcal{H}_x^{(\leq r)}(C)$ inside the subsets of $E \setminus \{x\}$ of size at most r. Its inclusion-minimal members form the minimal recovery-set clutter. Thus exact supports, recovery sets, and minimal recovery sets are distinct but determine the same monotone success event.

The coefficient layer is

$$\mathcal{P}_x^{(\leq r)}(C) = \{w \in C^\perp : w_x = 1, \operatorname{wt}(w) \leq r + 1\}.$$

These normalized recovery equations determine their exact supports, while the converse generally fails. The operational model is scalar exact repair: each participating helper supplies one symbol of $\mathbb{F}_q$. Array-code access, subpacketization, and repair bandwidth are different optimization problems [29, 31, 16].

If $A \subseteq E \setminus \{x\}$ is the surviving helper set, put

$$f_{x,r}(A) = 1 \quad \iff \quad A \text{ contains a member of } \mathcal{H}_x^{(\leq r)}(C).$$

Under independent helper survival with probabilities (s_y), the bounded repair reliability is

$$R_{x,r}((s_y)) = \mathbb{E}[f_{x,r}].$$

It depends only on the minimal recovery sets, but it is not determined by their minimum cardinality. Section 5 shows that even the complete relative-weight hierarchy does not determine the corresponding bounded reliability law.

For several target coordinates, a recovery equation is a dual subspace rather than one dual word. Let $P \subseteq E$, $J = E \setminus P$, and let $G = (G_P \mid G_J)$ be a full-row-rank generator matrix. If $T \leq \operatorname{im} G_P$ is a subspace of target linear combinations, a normalized recovery system for T consists of linear maps

$$\alpha : T \longrightarrow \mathbb{F}_q^P, \qquad \beta : T \longrightarrow \mathbb{F}_q^J$$

such that

$$G_P\alpha = \mathrm{id}_T, \qquad G_J\beta = -\,\mathrm{id}_T\,.$$

For each $t \in T$, the vector $(\alpha(t), \beta(t))$ is then a dual word. The map α records the chosen target normalization and β records the helper coefficients.

The helper union of the system is the support of $\beta(T)$:

$$\operatorname{supp}(\beta(T)) = \{j \in J : \text{some } t \in T \text{ has } \beta(t)_j \neq 0\}.$$

Its helper cost is $|\operatorname{supp}(\beta(T))|$. We call a target subspace internally recoverable when it lies in

$$W_P = \operatorname{im} G_P \cap \operatorname{im} G_J.$$

Only this recovered subspace dimension is counted below. Relations supported entirely on P have recovered dimension zero and do not consume helpers.

3 Puncturing, shortening, and relative recovery costs

Continue with $G = (G_P \mid G_J)$ and set

$$U_P = \operatorname{im} G_P, \qquad W_P = U_P \cap \operatorname{im} G_J,$$

$$K_P = \ker G_J, \qquad D_P = G_J^{-1}(U_P).$$

Proposition 3.1 (Associated nested code pair)**.** *Restriction of G_J gives an exact sequence*

$$0 \longrightarrow K_P \longrightarrow D_P \xrightarrow{G_J} W_P \longrightarrow 0. \tag{1}$$

Proof. By definition, $K_P = \ker G_J$ is contained in $D_P = G_J^{-1}(U_P)$. The kernel of the restricted map is therefore K_P. Its image is

$$G_J(D_P) = \operatorname{im} G_J \cap U_P = W_P,$$

so the restricted map is onto W_P. □

The first isomorphism theorem applied to the restricted map gives $D_P/K_P \cong W_P$.

Proposition 3.2 (Canonical puncturing–shortening description)**.** *With puncturing and shortening taken onto the helper set J,*

$$D_P = \mathrm{punct}_J(I^\perp), \qquad K_P = \mathrm{short}_J(I^\perp). \tag{2}$$

Taking duals inside $\mathbb{F}_q^J$ gives

$$D_P^\perp = \mathrm{short}_J(I), \qquad K_P^\perp = \mathrm{punct}_J(I). \tag{3}$$

Consequently the nested pair depends only on the code I and the split $E = P \sqcup J$; changing the row basis of a generator matrix leaves it unchanged.

Proof. A vector $a \in \mathbb{F}_q^J$ lies in $\mathrm{punct}_J(I^\perp)$ exactly when some $p \in \mathbb{F}_q^P$ satisfies $G_P p + G_J a = 0$, equivalently $G_J a \in \operatorname{im} G_P$, which is $a \in D_P$. It lies in $\mathrm{short}_J(I^\perp)$ exactly when $(0, a) \in I^\perp$, equivalently $G_J a = 0$, which is $a \in K_P$. The dual identities follow from the standard puncturing–shortening duality. □

Thus a monomial relabeling of helpers gives the corresponding monomially equivalent pair, while a row-basis change does nothing to the pair itself.

For a subspace $A \leq \mathbb{F}_q^J$, write $\operatorname{supp} A$ for the union of the supports of its vectors. For nested linear codes $K \subseteq D \leq \mathbb{F}_q^J$, the tth relative generalized Hamming weight is

$$M_t(D, K) = \min\{|\operatorname{supp} L| : L \leq D, \dim L - \dim(L \cap K) = t\}.$$

Equivalently, one may minimize over subspaces disjoint from K and of dimension t. Strict growth and the relative Singleton bound are Proposition 2 and Theorem 4 of Luo et al. [19]; standard treatments also cover ordinary Wei duality and computational forms [11, 28]. For nested code pairs in linear secret sharing, RDLPs and RGHWs already give exact information and reconstruction thresholds [15].

The following result is the standard nested-code support invariant applied to the canonical puncturing–shortening pair, with the normalization and helper support correspondence stated in recovery language.

Theorem 3.3 (Relative weights are exact recovery costs)**.** *Let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The minimum helper-union size among recovery systems for all internally recoverable t-dimensional subspaces of target linear combinations is*

$$\mu_t(P) = M_t(D_P, K_P).$$

More precisely, normalized recovery systems of recovered dimension t are in support-preserving correspondence, after forgetting the target normalization, with subspaces

$$L \leq D_P, \qquad \dim L = t, \qquad L \cap K_P = 0,$$

while the target normalization is a linear section through G_P.

Proof. Let a normalized system recover a t-dimensional subspace $T \leq W_P$. In the notation of Section 2, put $L = \beta(T)$. Since $G_J\beta = -\mathrm{id}_T$, the map β is injective, $L \leq D_P$, and $L \cap K_P = 0$. Moreover $G_J L = T$, and the union of helpers in the system is exactly $\operatorname{supp} L$.

Conversely, let $L \leq D_P$ have dimension t and meet K_P trivially. Then $G_J|_L$ is an isomorphism from L onto the t-dimensional subspace $T = G_J L \leq W_P$. Choose a linear section $\alpha : T \to \mathbb{F}_q^P$ with $G_P\alpha = \mathrm{id}_T$. Put $\beta = -(G_J|_L)^{-1}$. Then $G_P\alpha + G_J\beta = 0$, so (α, β) is a normalized recovery system with helper union $\operatorname{supp} L$.

It remains to compare this minimum with the standard relative-weight definition. If $L' \leq D_P$ satisfies

$$\dim L' - \dim(L' \cap K_P) = t,$$

choose a vector-space complement L to $L' \cap K_P$ inside L'. Then $\dim L = t$, $L \cap K_P = 0$, and $\operatorname{supp} L \subseteq \operatorname{supp} L'$. The reverse inclusion of feasible classes is immediate. The two minima are equal. □

The corresponding relative dimension/length profile is

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H|=s}} \big(\dim(D_P \cap \mathbb{F}_q^H) - \dim(K_P \cap \mathbb{F}_q^H)\big),$$

where $\mathbb{F}_q^H$ denotes vectors supported on H.

Proposition 3.4 (Recovery interpretation of the relative profile)**.** *For $0 \leq s \leq |J|$,*

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H|=s}} \dim\big(W_P \cap \operatorname{span}(G_H)\big).$$

Consequently,

$$M_t(D_P, K_P) = \min\{s : K_s(D_P, K_P) \geq t\}.$$

Thus K_s is the maximum number of independent target linear combinations recoverable from a fixed budget of s helpers.

Proof. Fix H. Restrict the exact map (1) to $D_P \cap \mathbb{F}_q^H$. Its kernel is $K_P \cap \mathbb{F}_q^H$, and its image is $W_P \cap \operatorname{span}(G_H)$. Rank–nullity gives the first equality after maximizing over H. The second is the standard inverse relation between the relative dimension/length profile and relative generalized weights; it also follows directly from Theorem 3.3. □

The shortening–puncturing duality behind the relative profiles of nested codes is standard in wiretap and ramp-secret-sharing theory [19, Lemma 1 and Theorem 2]; see also [11, eqs. (1)–(2)]. Its recovery interpretation gives the failure-side counterpart of Theorem 3.3.

Proposition 3.5 (Failures and residual ambiguity)**.** *Let $\ell = \dim W_P$. By (3), the dual nested pair is* $\operatorname{short}_J(I) \subseteq \operatorname{punct}_J(I)$*. If the helpers in $H \subseteq J$ survive and those in $F = J \setminus H$ fail, then the unrecovered target dimension is*

$$\ell - \dim\big(W_P \cap \operatorname{span}(G_H)\big) = \dim(K_P^{\perp} \cap \mathbb{F}_q^F) - \dim(D_P^{\perp} \cap \mathbb{F}_q^F). \tag{4}$$

Consequently, for $1 \leq t \leq \ell$,

$$M_t(K_P^{\perp}, D_P^{\perp}) = \min\left\{|F| : \dim\big(W_P \cap \operatorname{span}(G_{J\setminus F})\big) \leq \ell - t\right\}. \tag{5}$$

Thus the dual relative weight is the minimum number of helper failures that leave at least t target dimensions ambiguous.

Proof. For a code $C \le \mathbb{F}_q^J$, puncturing onto F gives

$$\dim C - \dim(C \cap \mathbb{F}_q^H) = \dim(C|_F) = |F| - \dim(C^\perp \cap \mathbb{F}_q^F).$$

Apply this identity to D_P and K_P and subtract. Since $\dim D_P - \dim K_P = \ell$, Proposition 3.4 gives (4). The right-hand side is the relative dimension of the dual pair $D_P^\perp \subseteq K_P^\perp$ on F. Its least support size at value at least t is $M_t(K_P^\perp, D_P^\perp)$: a subspace attaining the relative weight is supported on such an F, while any relative dimension at least t admits a t-dimensional subspace disjoint from $D_P^\perp \cap \mathbb{F}_q^F$. This proves (5). □

The strict inequalities

$$1 \le M_1(D_P, K_P) < \cdots < M_\ell(D_P, K_P)$$

therefore say that each additional independent target combination requires a strictly larger helper union. These minima solve the single-block problem but do not yet provide recursive state: the next section restores the functional labels needed for exact concatenation. Only after an outer-distance gate removes every nonzero label does the answer collapse to $M_t(D_P, K_P) + d(I^\perp)$.

4 Exact confinement under concatenation

The argument has two stages. First, every concatenated recovery system is classified by its outer functional label. The zero label gives a local recovery system plus an inner-dual perturbation; each nonzero label prescribes a coset in every active block. Minimizing these two sectors gives the exact finite nonconfinement cost. Second, the full label-indexed cost functions, rather than their global scalar minima, are passed to the next concatenation level by an associative min–sum law.

Let $L/\mathbb{F}_q$ be a finite extension and let $\iota : L \xrightarrow{\sim} I \le \mathbb{F}_q^E$ be an $\mathbb{F}_q$-linear isomorphism, used as the inner encoder. Assume $0 < \dim I < |E|$. Whenever the outer alphabet L occurs below, I denotes the represented inner code (I, ι), not only its image subspace. In particular, $O \circ I$, Φ_I, $\lambda_{I,T}$, $\mu_{I,P,T}$, and $\Gamma_{j,T}(O, I)$ depend on ι. The earlier pair $K_P \subseteq D_P$ depends only on the image code and target/helper split. Proposition 4.2 shows that this distinction cannot be suppressed under concatenation. For an L-linear outer code $O \le L^N$, write

$$O \circ I = \{(\iota(a_1), \ldots, \iota(a_N)) : (a_1, \ldots, a_N) \in O\}.$$

The proof of confinement uses the linear map

$$\Phi_I : \mathbb{F}_q^E \longrightarrow L^*, \qquad \Phi_I(y)(a) = \langle y, \iota(a) \rangle.$$

Here $L^* = \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$. Its kernel is $I^\perp$. A block vector $y = (y_1, \ldots, y_N)$ belongs to $(O \circ I)^\perp$ precisely when

$$(\Phi_I(y_1), \ldots, \Phi_I(y_N)) \in \mathrm{FD}(O), \tag{6}$$

where

$$\mathrm{FD}(O) = \{(\beta_j) \in (L^*)^N : \textstyle\sum_j \beta_j(a_j) = 0 \text{ for every } (a_j) \in O\}.$$

Indeed, both conditions say that y is orthogonal to every encoded outer word. Write $\beta_j(a) = \mathrm{Tr}_{L/\mathbb{F}_q}(b_j a)$. The functional-dual condition becomes

$$\mathrm{Tr}_{L/\mathbb{F}_q}\left(\sum_j b_j a_j\right) = 0 \qquad \text{for every } (a_j) \in O.$$

Because O is L-linear, the same holds after multiplying (a_j) by every $\lambda \in L$. Nondegeneracy of the trace pairing then forces $\sum_j b_j a_j = 0$. Hence $(b_j) \in O^\perp$, and the converse is immediate. This identification preserves block support, so the support distance of $\mathrm{FD}(O)$ is $d(O^\perp)$.

Fix a target block and a nonzero target-message subspace $T \leq U_P$. Let $\rho_T(I)$ be the minimum helper-union size of a normalized recovery system for T in the inner code, with value ∞ when no such system exists. Thus $\rho_T(I) < \infty$ exactly when $T \leq W_P$. In a concatenation, a normalized system for T means a linear family of dual equations whose target-block coefficient map $\alpha : T \to \mathbb{F}_q^P$ satisfies $G_P\alpha = \mathrm{id}_T$ under the fixed inner-message identification. This agrees with the intrinsic target image of the concatenated generator whenever the outer projection onto the chosen block is surjective. In particular, $d(O^\perp) > 1$ guarantees surjectivity: an L-linear coordinate image is either 0 or L, and a zero image would put a weight-one word in $O^\perp$. A system is *confined* if every one of its equations is supported in the target block.

For an $\mathbb{F}_q$-linear map $B : T \to L^*$, define

$$\lambda_{I,T}(B) = \min_{\substack{Y:T\to\mathbb{F}_q^E \text{ linear} \\ \Phi_I Y = B}} |\operatorname{supp} Y(T)|. \tag{7}$$

For the target block, define

$$\mu_{I,P,T}(B) = \min_{\substack{\alpha:T\to\mathbb{F}_q^P,\ \beta:T\to\mathbb{F}_q^J \text{ linear} \\ G_P\alpha=\mathrm{id}_T \\ \Phi_I(\alpha,\beta)=B}} |\operatorname{supp} \beta(T)|. \tag{8}$$

An empty minimum is ∞. After choosing a basis of T, (7) is the minimum union support of representatives of prescribed cosets of $I^\perp$. This is the fixed-instance quantity underlying generalized coset weights and generalized covering radii [32, 8]. The target version retains the normalization required by the recovery problem. In particular,

$$\mu_{I,P,T}(0) = \rho_T(I). \tag{9}$$

Let

$$\mathcal{B}_T(O) = \operatorname{Hom}_{\mathbb{F}_q}\big(T, \mathrm{FD}(O)\big).$$

For $B \in \mathcal{B}_T(O)$, write $B = (B_h)_{h=1}^N$ with $B_h : T \to L^*$. Assume $N \geq 2$, fix a target block j whose coordinate projection $O \to L$ is nonzero, hence surjective, and put

$$\Gamma_{j,T}(O,I) = \min\Bigg\{\mu_{I,P,T}(0) + d(I^\perp), \quad \min_{0\neq B\in\mathcal{B}_T(O)} \Bigg(\mu_{I,P,T}(B_j) + \sum_{h\neq j} \lambda_{I,T}(B_h)\Bigg)\Bigg\}. \tag{10}$$

Theorem 4.1 (Exact nonconfinement cost from labelled coset supports)**.** *Under the hypotheses above, $\Gamma_{j,T}(O,I)$ is the least helper-union size of a nonconfined normalized recovery system for T in $O \circ I$, with value ∞ if no such system exists. Thus every cost-$\leq r$ system for T is confined if and only if*

$$r < \Gamma_{j,T}(O,I). \tag{11}$$

For $1 \leq t \leq \ell = \dim W_P$, set

$$\Gamma_{j,t}(O,I) = \min_{\substack{T\leq W_P \\ \dim T = t}} \Gamma_{j,T}(O,I). \tag{12}$$

Then every cost-$\leq r$ system for every internally recoverable t-dimensional target subspace is confined if and only if $r < \Gamma_{j,t}(O, I)$. Below either threshold, restriction and zero-extension are inverse bijections on normalized systems and preserve exact helper supports.

Proof. Write a recovery system as an $\mathbb{F}_q$-linear map $Y : T \to (O \circ I)^\perp$ and let Y_h be its block maps. By (6),

$$B_h = \Phi_I Y_h$$

defines a linear map $B : T \to \mathrm{FD}(O)$. Conversely, compatible linear lifts of the block maps of any such B give a map into the concatenated dual. At the target block, compatibility with the chosen target space is exactly the constraint in (8).

Suppose first that $B \neq 0$. No nonzero element of $\mathrm{FD}(O)$ is supported only at j: testing such an element against outer words whose jth coordinate ranges over L would force its jth functional to vanish. Hence some block map B_h with $h \neq j$ is nonzero, and every lift is nonconfined. Distinct inner blocks have disjoint helper coordinates, so the minimum helper union for this fixed B is

$$\mu_{I,P,T}(B_j) + \sum_{h \neq j} \lambda_{I,T}(B_h).$$

The block lifts can be chosen independently, so this lower bound is attained.

If $B = 0$, every block map takes values in $I^\perp = \ker \Phi_I$. The target block costs $\rho_T(I) = \mu_{I,P,T}(0)$. Nonconfinement requires a nonzero map $T \to I^\perp$ in another block. Its image has union support at least $d(I^\perp)$, and equality is attained by $u \mapsto f(u)v$, where $0 \neq f \in T^*$ and v is a minimum-weight word of $I^\perp$.

The zero and nonzero functional sectors are exhaustive, proving (10) and (11). Minimizing over t-dimensional T proves the ranked statement. Below the first nonconfined cost, restriction to block j is inverse to zero-extension, with coefficients and supports unchanged. □

A complete small instance. Let $I \leq \mathbb{F}_2^3$ have generator columns $G = (e_1 \mid e_2, e_1 + e_2)$, with the first coordinate targeted. Then $K_P = 0$, $D_P = \langle(1, 1)\rangle$, and $M_1(D_P, K_P) = 2$: the target is recovered from exactly the two helpers. Moreover $I^\perp = \langle(1, 1, 1)\rangle$, so the zero-functional escape cost is $2 + 3 = 5$.

Identify the two-dimensional message space with $L = \mathbb{F}_4$ and concatenate with the length-six single-parity-check code over L. Its functional dual is the repetition line. Every nonzero label is active in all five blocks outside the target block, and each active block costs at least one helper. Thus its nonzero-functional sector also costs at least five, so $\Gamma_{j,1} = 5$. Every recovery system of cost at most four is therefore confined, while a cost-five witness is obtained by adjoining the inner-dual word $(1, 1, 1)$ in another block. The same copied support becomes a scheduler load vector: two demands in distinct blocks fit unit helper capacities, whereas two demands in one block compete for the same helper pair. This example will serve as the model case for the composition and scheduling constructions.

4.1 Repeated concatenation

The ordinary prescribed-coset costs retain the intermediate labels needed at the next concatenation level. Target-normalized composition also retains the target image in each inner block. The resulting formulas are instances of variable elimination over the min-sum semiring [2]. This is also the algebraic form of passing symbol-indexed inner costs to an outer decoder, as in soft and generalized-concatenated decoding [13, 3].

The zero-functional escape cost alone is not compositional. The next separation holds the underlying binary code and all its unlabelled recovery data fixed.

Proposition 4.2 (Functional labels are necessary for finite composition)**.** *Over $\mathbb{F}_2 \subseteq \mathbb{F}_4$, there are two represented inner encoders with the same image code, dual code, target/helper nested pair, complete relative-weight hierarchy, dual distance, unlabelled normalized helper-support family, and zero-functional escape costs for every recoverable target subspace, but one fixed outer represented code gives different exact nonconfinement costs.*

Proof. Write the nonzero elements of $\mathbb{F}_4$ as $1, \omega, \omega^2 = 1 + \omega$ and take generator columns

$$I_1 : (1, 1, \omega), \qquad I_2 : (1, 1, \omega^2),$$

with the first coordinate targeted. Both generator matrices have binary row space

$$\{(x, x, y) : x, y \in \mathbb{F}_2\}$$

and dual $\langle(1, 1, 0)\rangle$. Thus both have dual distance 2, $K_P = 0 \subseteq D_P = \langle(1, 0)\rangle$, $M_1 = 1$, and the same unique minimal helper support. Their recoverable target space is one-dimensional, so their complete zero-functional profile is the common value $1 + 2 = 3$.

On $(1, \omega, \omega^2)$, their ordinary coset costs are $(1, 1, 2)$ and $(1, 2, 1)$, while their normalized target costs are $(0, 2, 1)$ and $(0, 1, 2)$. Take the fixed outer code whose functional dual is the line spanned by $(1, \omega)$. The three nonzero scalar multiples have respective target-plus-external costs

$$(1, 4, 2) \qquad \text{and} \qquad (2, 2, 3).$$

Equation (10) therefore gives exact costs 1 and 2. Only the alignment of costs with the intermediate functional labels has changed. □

Let $\mathbb{F}_q \subseteq L \subseteq M$ be a tower of finite fields. Let B be an $\mathbb{F}_q$-linear represented code with message space L, and let A be an L-linear represented code with message space M. After the trace identifications, write their dual restriction maps as

$$\phi_B : \mathbb{F}_q^{E_B} \longrightarrow L, \qquad \phi_A : L^{E_A} \longrightarrow M.$$

For an $\mathbb{F}_q$-space T and $b : T \to L$, put

$$\Lambda_{B,T}(b) = \min_{\phi_B Y = b} |\operatorname{supp} Y(T)|.$$

Define $\Lambda_{A,T}$ and $\Lambda_{A\circ B,T}$ similarly. If $J \subseteq E_B$, let $\Lambda_{B,J,T}$ denote the same minimum with Y supported on J and ϕ_B restricted to those coordinates.

Theorem 4.3 (Min–sum closure of prescribed-coset support costs)**.** *For every $\mathbb{F}_q$-linear map $c : T \to M$,*

$$\Lambda_{A\circ B,T}(c) = \min_{X:T\to L^{E_A}\ \phi_A X = c} \sum_{e\in E_A} \Lambda_{B,T}(X_e). \tag{13}$$

When $T \neq 0$, set

$$\delta_{B,T} = \min_{0\neq b:T\to L} \Lambda_{B,T}(b), \qquad R_{B,T} = \max_{0\neq b:T\to L} \Lambda_{B,T}(b).$$

For $c \neq 0$, the exact formula has the sharp coarsening

$$\delta_{B,T}\Lambda_{A,T}(c) \le \Lambda_{A\circ B,T}(c) \le R_{B,T}\Lambda_{A,T}(c). \tag{14}$$

Neither constant can be improved without further hypotheses. Suppose that a target set in $A \circ B$ meets block e in $P_e \subseteq E_B$, and put $J_e = E_B \setminus P_e$ and $U_{P_e} = \operatorname{im} \phi_{B,P_e}$. If $T \leq M$ and $\iota_T : T \hookrightarrow M$ is the inclusion, then the target-normalized cost satisfies

$$\mu_{A\circ B,P,T}(c) = \min_{\substack{U,X:T\to L^{E_A} \\ U_e(T)\subseteq U_{P_e}\ (e\in E_A) \\ \phi_A U=\iota_T,\ \phi_A X=c}} \sum_{e\in E_A} \Lambda_{B,J_e,T}(X_e - U_e). \tag{15}$$

Moreover,

$$d((A\circ B)^\perp) = \min\left\{ d(B^\perp), \min_{0\neq z\in A^\perp} \sum_{e\in E_A} \Lambda_{B,\mathbb{F}_q}(u \mapsto u z_e) \right\}. \tag{16}$$

These formulas iterate associatively. Substitution into (10) therefore gives the exact nonconfinement cost through any finite tower, with the zero and nonzero functional sectors kept separate at every level.

Proof. A leaf-level lift $Y : T \to \mathbb{F}_q^{E_A\times E_B}$ determines intermediate maps $X_e = \phi_B Y_e$. Its support is the disjoint union of its supports in the inner blocks. At fixed X, the block lifts are independent and attain their minima separately, which proves (13). Every nonzero intermediate coordinate map costs between $\delta_{B,T}$ and $R_{B,T}$. Minimizing its number of nonzero coordinates proves (14); a one-coordinate outer realization and a map attaining either extremum show that the constants are sharp.

For the target formula, let $a_e : T \to \mathbb{F}_q^{P_e}$ be the target coefficients and set $U_e = \phi_{B,P_e} a_e$. Every map into U_{P_e} has such a linear lift. Target normalization is exactly $\phi_A U = \iota_T$. If X_e is the total intermediate map in block e, its helper contribution must lift $X_e - U_e$ through ϕ_{B,J_e}, proving (15). Before taking minimum supports, retain the actual target maps a_e and the full helper lift sets. These compose by fibre product over the intermediate total maps X, so the same argument also transports every coefficient-labelled equation and its support.

A word in $(A\circ B)^\perp$ either has zero intermediate label, when a minimum word of $B^\perp$ in one block is optimal, or has a nonzero label $z \in A^\perp$, when the block representatives minimize independently. This proves (16). At three or more levels, either parenthesization minimizes over the same intermediate arrays and sums the same leaf-block costs, proving associativity and the final assertion. □

The trace identifications in this proposition are compatible with the tower. Indeed, trace transitivity gives

$$\sum_e \operatorname{Tr}_{L/\mathbb{F}_q}\big(X_e(v)\,\iota_A(m)_e\big) = \operatorname{Tr}_{M/\mathbb{F}_q}\big(\phi_A(X(v))m\big),$$

and nondegeneracy of the trace pairing identifies the composite restriction map with $\phi_A \circ \phi_B^{E_A}$, where $\phi_B^{E_A}$ applies ϕ_B blockwise.

For rank-one targets, the exact quotient of the labelled functions observable by outer concatenation has a projective description. Fix a line $T = \langle u\rangle \leq W_P$. For $b \in L$, let $\tau_b : T \to L^*$ be defined by

$$\tau_b(au)(x) = a \operatorname{Tr}_{L/\mathbb{F}_q}(bx) \qquad (a \in \mathbb{F}_q,\ x \in L),$$

and put $z_{I,T} = \rho_T(I) + d(I^\perp)$. If $[b] = [b_1 : \cdots : b_N] \in \mathbf{P}(L^N)$ has a nonzero coordinate outside the target block j, define

$$\Theta_{I,T}([b]) = \min_{s\in L^\times} \left(\mu_{I,P,T}(\tau_{sb_j}) + \sum_{h\neq j} \lambda_{I,T}(\tau_{sb_h}) \right). \tag{17}$$

Changing u or the projective representative of $[b]$ does not change this value.

Here a *rank-one outer context* is a triple (N, j, O) with $N \geq 2$, $O \leq L^N$ an L-linear code, and nonzero projection onto the distinguished block j. The context family ranges over every such N and j, while the inner target/helper split and the identified target line T remain fixed. Two represented inner codes are *contextually equivalent at* T when their values of $\Gamma_{j,T}$ agree in every member of this family. A compatible further concatenation adds outer layers above the same distinguished leaf and does not retarget that leaf.

Theorem 4.4 (Minimal contextual quotient at rank one)**.** *Let $D = O^{\perp} \leq L^N$ under the trace identification, and suppose the projection of O onto block j is nonzero. Then*

$$\Gamma_{j,T}(O, I) = \min\left\{ z_{I,T}, \min_{\substack{Lb \leq D \\ b \neq 0}} \Theta_{I,T}([b]) \right\}, \tag{18}$$

where the inner minimum is empty when $D = 0$.

Consequently, represented inner codes over the same extension field with identified target lines are contextually equivalent at T if and only if their values $z_{I,T}$ agree and the family of truncated profiles

$$[b] \longmapsto \min\{z_{I,T}, \Theta_{I,T}([b])\}$$

agrees for every $N \geq 2$, target block j, and projective tuple with nonzero external part. Thus equality of this response family realizes the exact numerical contextual quotient at rank one: any invariant determining those costs in every outer context must refine its equivalence classes. It is enough to test the full outer code and outer codes whose functional dual has L-dimension one.

Proof. A nonzero map from the one-dimensional $\mathbb{F}_q$-space T into D is determined by one nonzero tuple $b \in D$. Partition $D \backslash \{0\}$ into its L-lines in the nonzero-functional sector of (10). Minimization on the line Lb is exactly (17), which proves (18).

It remains to show that every stated probe is an actual outer-code context. The full outer code $O = L^N$ has $D = 0$ and exposes $z_{I,T}$. Given a projective tuple $[b]$ with nonzero external part, take $D = Lb$ and $O = D^{\perp}$. Its target projection is nonzero: otherwise D would contain the target coordinate line, forcing every external coordinate of b to vanish. This context exposes $\min\{z_{I,T}, \Theta_{I,T}([b])\}$. Equality in every context therefore forces equality of the displayed response family; the first part of the theorem gives the converse.

For the congruence statement, place the same distinguished target leaf in two composites $A \circ I_1$ and $A \circ I_2$. Any compatible later context C tests the target-normalized composites $C \circ (A \circ I_i) = (C \circ A) \circ I_i$. The target-normalized composition formula (15) carries the same leaf target image on both parenthesizations, and $C \circ A$ is one of the contexts quantified above. Contextual equivalence of I_1 and I_2 therefore gives equal costs after every such C. Hence contextual equivalence is a congruence for the specified compatible concatenations. This is a numerical statement; full coefficient witnesses still require the minimizing lifts described after Theorem 4.3. $\square$

The same generated-image argument bounds the dimension of the outer contexts needed at every target rank.

Proposition 4.5 (Rank-bounded outer tests)**.** *Let $\dim_{\mathbb{F}_q} T = t$, put $D = O^{\perp}$, and retain the nonzero projection onto block j. For L-subspaces $D' \leq D$, set $O_{D'} = (D')^{\perp} \leq L^N$. Then*

$$\Gamma_{j,T}(O, I) = \min_{\substack{D' \leq D \\ \dim_L D' \leq t}} \Gamma_{j,T}(O_{D'}, I). \tag{19}$$

Thus outer codes with functional-dual dimension at most t form a complete test family for exact contextual equivalence at target rank t.

Proof. For $B : T \to D$, let $D_B = \operatorname{span}_L B(T)$. The image of an $\mathbb{F}_q$-basis of T generates D_B over L, so $\dim_L D_B \leq t$. Regarding B as a map into D_B changes none of its block labels or costs. Conversely, every map into $D' \leq D$ is a map into D. Minimizing the same cost functional in both directions proves (19); $D' = 0$ supplies the zero-functional sector. No $D' \leq D$ contains a nonzero target-supported vector, so every $O_{D'}$ retains nonzero target projection. ☐

The simultaneous confinement gate across all recoverable ranks has a single bottleneck.

Corollary 4.6 (Rank-one criterion for complete bounded transfer)**.** *For every target block j with nonzero outer projection,*

$$\min_{0 \neq T \leq W_P} \Gamma_{j,T}(O, I) = \Gamma_{j,1}(O, I). \tag{20}$$

For every integer $r \geq 0$, the following are equivalent:

1. *every cost-$\leq r$ system on every internally recoverable target line is confined;*
2. *every cost-$\leq r$ system on every nonzero internally recoverable target subspace is confined; and*
3. *$r < \Gamma_{j,1}(O, I)$.*

Under these conditions, restriction and zero-extension are inverse bijections simultaneously for every nonzero $T \leq W_P$, preserving every coefficient and exact helper support through radius r. Hence they preserve all bounded reliability laws and every support- or coefficient-determined capacity and scheduling region on these systems. They also preserve every targetwise and rankwise minimum helper-union cost after truncation at $r+1$; when all relevant inner minima are at most r, the untruncated minima agree.

Proof. If a system on $0 \neq T \leq W_P$ is nonconfined, some external block map is nonzero on a vector $u \in T$. Restriction to $\langle u \rangle$ remains nonconfined and cannot enlarge its helper union. This proves the displayed equality, and Theorem 4.1 gives the three equivalent gates. Below the gate every bounded system is confined, so restriction deletes only zero external blocks and zero-extension is its inverse. The coefficient, support, reliability, capacity, and scheduling assertions follow from this bijection. For a minimum helper-union cost, either the inner minimum is at most r and the bijection gives equality, or both inner and concatenated minima exceed r; this is precisely equality after truncation at $r+1$. ☐

The exact optimization also recovers the outer-distance collapse without choosing a radius. Indeed, every nonzero functional sector costs at least $d(O^\perp) - 1$, while the zero-functional sector attains $M_t(D_P, K_P) + d(I^\perp)$ after minimization over T. Consequently,

$$d(O^\perp) \geq M_t(D_P, K_P) + d(I^\perp) + 1 \implies \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^\perp). \tag{21}$$

Theorem 4.7 (Confinement for a fixed target subspace)**.** *Let r be a nonnegative integer and let $O \leq L^N$ be an L-linear outer code with $N \geq 2$ and $d(O^\perp) > r + 1$, and assume $T \leq W_P$. Then $O \circ I$ confines every normalized recovery system for T of cost at most r if and only if*

$$r < \rho_T(I) + d(I^\perp). \tag{22}$$

Below this bound, zero-extension gives a support-preserving bijection between the inner systems of cost at most r and the corresponding systems in the concatenation. Hence the same conclusions hold for all sufficiently long members of any outer family with dual distance tending to infinity, at every fixed r.

Proof. If $0 \neq B \in \mathcal{B}_T(O)$, choose $u \in T$ with $B(u) \neq 0$. The tuple $B(u) \in \mathrm{FD}(O)$ has at least $d(O^\perp)$ nonzero blocks. At most one is the target block, so every lift has at least $d(O^\perp) - 1 > r$ helpers outside it. Thus no nonzero-functional sector in (10) occurs at cost at most r.

The zero-functional sector has exact cost $\mu_{I,P,T}(0) + d(I^\perp) = \rho_T(I) + d(I^\perp)$. Theorem 4.1 now gives both directions of (22) and the support-preserving bijection. □

Minimizing the fixed-subspace threshold over all recovered subspaces of fixed dimension gives the promised hierarchy.

Theorem 4.8 (Confinement by recovered dimension after the outer-distance gate)**.** *Let r be a nonnegative integer, let $O \leq L^N$ be L-linear with $N \geq 2$ and $d(O^\perp) > r + 1$, and let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The concatenation $O \circ I$ confines every cost-$\leq r$ recovery system for every internally recoverable t-dimensional target subspace if and only if*

$$r < M_t(D_P, K_P) + d(I^\perp). \tag{23}$$

When (23) holds, restriction and zero-extension preserve every normalized equation and its exact helper support. For an outer family with dual distance tending to infinity, these conclusions hold for all sufficiently long members at every fixed r.

Proof. By Theorem 3.3, every such target subspace has inner cost at least $M_t(D_P, K_P)$, and some target subspace attains this minimum. Apply Theorem 4.7 to each subspace. The lower bound gives sufficiency uniformly; an attaining subspace and the external rank-one perturbation in that proof give necessity. □

4.2 One-coordinate specialization

For $\beta \in L^*$, put

$$\lambda_I(\beta) = \min_{\Phi_I(y)=\beta} \mathrm{wt}(y), \qquad \mu_{I,x}(\beta) = \min_{\substack{\Phi_I(y)=\beta \\ y_x \neq 0}} \mathrm{wt}(y),$$

with value ∞ when a constrained fibre is empty. The first quantity is the usual coset-leader weight for the inner encoder, while the second retains the target-coordinate constraint. The functional-dual decomposition itself is classical concatenation machinery [6, Theorem 2.1].

Theorem 4.9 (Exact weighted finite-length confinement at one coordinate)**.** *Fix an outer block j, an inner coordinate x, and $N \geq 2$, and assume that the coordinate projection $O \to L$ at block j is nonzero, hence surjective. The minimum total Hamming weight of a word in $(O \circ I)^\perp$ that is nonzero at (j, x) and is not the zero-extension of a word of $I^\perp$ in block j is*

$$Z_{j,x}(O, I) = \min \left\{ \mu_{I,x}(0) + d(I^\perp), \min_{\substack{\beta=(\beta_h)\in\mathrm{FD}(O) \\ \beta \neq 0}} \left(\mu_{I,x}(\beta_j) + \sum_{h \neq j} \lambda_I(\beta_h) \right) \right\}. \tag{24}$$

A minimum over an empty set is ∞. Consequently, for every $r \geq 0$, restriction to block j and zero-extension are inverse, support-preserving bijections between the normalized recovery equations through x in I and those through (j, x) in $O \circ I$, both using at most r helpers, if and only if

$$Z_{j,x}(O, I) > r + 1. \tag{25}$$

The same equivalence holds for the exact helper-support families. When these conditions hold, the inclusion-minimal recovery sets are copied as well.

Proof. Write a concatenated dual word in blocks and apply (6). For a fixed nonzero functional tuple, the block representatives minimize independently, giving the second term of (24). For the zero tuple, nonconfinement costs $\mu_{I,x}(0)$ in the target block and $d(I^{\perp})$ in another block. The projection hypothesis makes confined words exactly the zero-extensions of inner-dual words, so these two sectors prove (24).

If the target column is nonzero, take $P = \{x\}$ and let $T \le U_P$ be its span. After choosing a nonzero vector of T, a linear map $T \to L^*$ is one functional. Normalizing its target coefficient and rescaling the functional tuple identify the minima in (10) with those in (24). The target coordinate contributes one to total word weight and is not a helper, so

$$Z_{j,x}(O, I) = \Gamma_{j,T}(O, I) + 1.$$

Theorem 4.1 gives the claimed gate. If the target column is zero, the same gate follows directly from the exact minimum already proved. In either case, below the minimum every equation is confined, so restriction and zero-extension are inverse. In one dimension every exact helper support is the support of its normalized equation; the equivalence also holds for exact helper-support families and their inclusion-minimal members. □

Theorem 4.9 retains information that the outer support distance discards. The following family makes the difference strict without using a geometric inner code.

Proposition 4.10 (A family from a Singer cycle beyond the outer-distance gate)**.** *Let $k \ge 2$ and suppose*

$$\frac{q^k - 1}{q - 1} > (k+1)^2. \tag{26}$$

Let I be the $[k+1, k, 2]_q$ single-parity-check code generated by

$$G = (I_k \mid \mathbf{1}),$$

and identify its message space with $L = \mathbb{F}_{q^k}$. For every $k+1 \le n \le 2k+1$, there is an L-linear length-n outer code O with $d(O^{\perp}) = n$ such that (25) holds at helper radius $n-1$ for every inner coordinate and every outer block. Thus all normalized radius-$(n-1)$ recovery equations and their exact helper supports are copied blockwise although the ordinary outer-distance gate fails.

In particular, this holds for the $[4, 3, 2]_5$ inner code and a length-five outer code at helper radius four.

Proof. The $k+1$ coordinate functionals of the encoder give a set S of $k+1$ points in the projective space on L^*. The multiplication maps of L, modulo $\mathbb{F}_q^*$, induce on that projective space a Singer cycle of order $(q^k-1)/(q-1)$, acting regularly on its points [30]. At most $|S|^2 = (k+1)^2$ elements of the cycle send a point of S to a point of S, so (26) permits a multiplication map $a : L \to L$, explicitly $a(u) = \gamma u$ for some $\gamma \in L^{\times}$, whose dual action satisfies $a^*(S) \cap S = \varnothing$.

Set

$$O = \{u \in L^n : u_0 + a(u_1) + u_2 + \cdots + u_{n-1} = 0\}.$$

Every functional-dual tuple has the form $(f, f \circ a, f, \ldots, f)$. If $f \ne 0$, all n entries are nonzero, so $d(O^{\perp}) = n$. A realization of total weight n would have weight one in every block. Weight-one realizations are exactly the scalar multiples of the $k+1$ coordinate functionals, so both $[f]$ and $[f \circ a]$ would lie in S, contrary to the choice of a. Every nonzero tuple therefore costs at least $n+1$.

Every outer coordinate projection is surjective. Finally, $I^{\perp}$ is spanned by $(1, \ldots, 1, -1)$, so $d(I^{\perp}) = \mu_{I,x}(0) = k+1$ for every x. The zero-functional cost in (24) is $2k+2 > n$, and the nonzero-functional cost is at least $n+1$. Thus $Z_{j,x}(O,I) > n$ for every j,x. At $r = n-1$ the ordinary gate would require $d(O^{\perp}) > n$, which is false, while the exact weighted gate holds. Taking $(q,k,n) = (5,3,5)$ gives the stated concrete instance. Since the inner recovery equations have k helpers and $n-1 \geq k$, the asserted transfer is nonvacuous throughout the family. □

For a nondegenerate singleton target $P = \{x\}$, the first relative weight is the least number of helpers in a dual word through x. If the total-weight threshold used for one-coordinate equations is denoted by $z_x(I)$, then

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^{\perp}). \tag{27}$$

The helper-radius condition (23) is therefore $r + 1 < z_x(I)$, with no change of convention hidden in the reduction. Theorem 4.1 is the common finite formula. The outer-distance gate removes its nonzero-functional sector and gives the RGHW threshold in every recovered dimension. Restricting instead to one target coordinate gives Theorem 4.9 without the gate.

The labelled functions are sufficient for exact numerical composition; minimizing lifts must also be stored to propagate witnesses. Their coarsenings answer different questions. The next section shows sharply that relative-weight minima do not determine recovery reliability and that the abstract nested pair does not determine confinement.

5 Sharp limits of coarser recovery invariants

The preceding section supplies an exact numerical state for composition. We now rule out two tempting coarsenings. Relative generalized weights record minimum support sizes but not the overlap pattern of all recovery sets, which controls stochastic repair. The abstract nested pair records more than those minima, but not its ambient inner-code realization; that presentation can change $d(I^{\perp})$ and the prescribed-coset support costs in (10). The support-side loss is

$$\text{normalized recovery systems} \longrightarrow \text{exact helper supports} \longrightarrow \text{relative-weight minima.}$$

The exact finite concatenation formula instead uses the outer functional dual and the joint support costs of prescribed inner cosets. The first separation below rules out the complete RGHW hierarchy as a reliability invariant at every quotient rank. The second rules out even the full abstract pair, together with the same unlabelled helper supports, as a confinement invariant.

First consider a one-dimensional recovered space. On the helper set $\{0,\ldots,8\}$, let

$$\mathcal{A} = \{158, 026, 045, 013, 478\},$$
$$\mathcal{B} = \{168, 237, 078, 015, 124\},$$

where, for example, 158 denotes $\{1,5,8\}$. Both are realizable as the minimum recovery sets at a distinguished coordinate of a represented $[10,4,6]_q$ code over a common finite prime field.

Proposition 5.1 (Rank-one support-overlap separation)**.** *The two distinguished coordinates above have the same minimum helper cost $M_1 = 3$. Under independent homogeneous helper survival with probability s, their radius-three repair reliabilities are*

$$R_{\mathcal{A}}(s) = 5s^3 - 7s^5 - s^6 + 5s^7 - s^9, \tag{28}$$
$$R_{\mathcal{B}}(s) = 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8. \tag{29}$$

In particular, the complete rank-one relative-weight hierarchy does not determine bounded repair reliability.

Proof. The five triples in each clutter are the minimum recovery sets, so both minimum costs are three. For $k = 1, \dots, 5$, the numbers of k-edge subfamilies according to union size are

k	$\mathcal{A}$	$\mathcal{B}$
1	$5u^3$	$5u^3$
2	$7u^5 + 3u^6$	$7u^5 + 3u^6$
3	$2u^6 + 6u^7 + 2u^8$	$u^6 + 8u^7 + u^8$
4	$u^7 + 2u^8 + 2u^9$	$4u^8 + u^9$
5	u^9	u^9.

Inclusion–exclusion for the five edge-containment events gives (28) and (29).

We now give the representation argument. Over an infinite field K, choose five projective lines $\ell_1, \dots, \ell_5$. For $\mathcal{A}$, make ℓ_2, ℓ_3, ℓ_4 concurrent and label

$$0 = \ell_2 \cap \ell_3 \cap \ell_4, \quad 1 = \ell_1 \cap \ell_4, \quad 4 = \ell_3 \cap \ell_5, \quad 5 = \ell_1 \cap \ell_3, \quad 8 = \ell_1 \cap \ell_5.$$

Choose $2, 6$ on ℓ_2, 3 on ℓ_4, and 7 on ℓ_5. For $\mathcal{B}$, make ℓ_1, ℓ_4, ℓ_5 concurrent at 1, put

$$8 = \ell_1 \cap \ell_3, \quad 7 = \ell_2 \cap \ell_3, \quad 0 = \ell_3 \cap \ell_4, \quad 2 = \ell_2 \cap \ell_5,$$

and choose $6, 3, 5, 4$ on $\ell_1, \ell_2, \ell_4, \ell_5$, respectively. At each free choice, avoid the finitely many unwanted joining lines. The displayed triples are then exactly the collinear triples.

Choose representatives $p_i \in K^3$ of either plane configuration, let $x = (1, 0, 0, 0)$, and lift helper i to $h_i = (t_i, p_i) \in K^4$. Dependence of the lifts over any displayed line and dependence of any four helper lifts are proper linear conditions on the t_i. For a displayed line, its three quotient vectors have a unique dependence up to scalar, and the same coefficients give one nonzero linear equation in the t_i. For four helper points, the quotient vectors span K^3, so their lifted determinant is a nonzero linear form in the four first coordinates. Avoiding this finite union makes every helper triple and quadruple independent, while the dependent four-sets through x are exactly $\{x\} \cup A$ for the five prescribed triples A. Choose the construction over $\mathbb{Q}$ and reduce both configurations modulo one prime avoiding the finitely many nonzero determinants. The two row-span codes then have rank four, dual distance four, and minimum recovery sets exactly $\mathcal{A}$ and $\mathcal{B}$. A hyperplane not containing x contains at most three helpers because every four helpers are independent. A hyperplane containing x contains at most one displayed collinear triple, and such a triple gives four columns in a hyperplane. Thus the maximum hyperplane intersection has size four, so both codes have distance $10 - 4 = 6$. $\square$

The rank-one separation extends to every quotient rank without changing any relative generalized weight. For a nested pair $K \subseteq D \leq \mathbb{F}_q^J$ with $\dim(D/K) = \ell$, define its radius-R full-quotient reliability by

$$\Pr\{\exists L \leq D : \ L + K = D, \ |\operatorname{supp} L| \leq R, \ \operatorname{supp} L \subseteq S\},$$

where $S \subseteq J$ is the random set of independently surviving helpers, each present with probability s.

Theorem 5.2 (The complete RGHW hierarchy does not determine reliability)**.** *For every $\ell \geq 1$, there are two nested code pairs of quotient dimension ℓ with identical relative generalized Hamming weights $M_1, \dots, M_\ell$ but different bounded reliability laws for recovery of the full quotient.*

Proof. For $\ell = 1$, use Proposition 5.1. For $\ell > 1$, let $K_0 \subset D_0$ be either of its two associated rank-one pairs. On disjoint helper blocks, add the same $\ell - 1$ pairs

$$0 \subset \langle \mathbf{1}_{L_i} \rangle \qquad (1 \leq i \leq \ell - 1),$$

where each padding line has the unique support of size L_i. Thus

$$K = K_0 \oplus 0 \oplus \cdots \oplus 0, \qquad D = D_0 \oplus \langle \mathbf{1}_{L_1} \rangle \oplus \cdots \oplus \langle \mathbf{1}_{L_{\ell-1}} \rangle.$$

For a direct sum of rank-one quotient pairs, the tth relative weight is the sum of the t smallest component costs. Both constructions therefore have the same hierarchy, obtained from the common multiset $\{3, L_1, \ldots, L_{\ell-1}\}$.

To verify the direct-sum formula, project a feasible helper subspace to each quotient component. A t-dimensional quotient image has nonzero projection to at least t of the one-dimensional components. On disjoint helper blocks, each active projection pays at least that component's minimum cost, so the total support is at least the sum of the t smallest costs. The direct sum of minimum words from the t cheapest components attains the bound.

Set $R = 3 + \sum_i L_i$. A recovery subspace of full quotient rank projects nontrivially to every padding component and hence uses its entire forced support. Removing those supports leaves a radius-three recovery set in the base component. Conversely, any base radius-three recovery set together with the padding generators recovers the full quotient at radius R. Thus

$$R_{\mathrm{full},R}(s) = s^{\sum_i L_i} R_{\mathrm{base},3}(s).$$

The two polynomials remain different by (28)–(29).

Every pair in the construction is realized by an inner code. Put $V = \mathbb{F}_q^J/K$, let $\pi : \mathbb{F}_q^J \to V$ be the quotient map, and choose an isomorphism $\alpha : \mathbb{F}_q^\ell \to D/K \le V$. The generator map

$$G : \mathbb{F}_q^\ell \oplus \mathbb{F}_q^J \longrightarrow V, \qquad G(a, x) = \alpha(a) + \pi(x),$$

has helper kernel K and helper preimage of the target span equal to D. Its associated nested pair is therefore $K \subseteq D$. □

The second separation fixes the abstract nested pair itself. This is not a row-basis change of one inner code: by Proposition 3.2, such a change leaves (D_P, K_P) unchanged. Here a *coefficient presentation* means an ambient inner dual code, target coordinates, and target coefficient map that realize a prescribed abstract pair $K \subseteq D$ as its shortening–puncturing pair. Different such realizations may have different ambient dual distances. Let

$$K = 0, \qquad D = \langle u, v \rangle \le \mathbb{F}_2^3, \qquad u = 100, \quad v = 011.$$

The nonzero helper words have weights $1, 2, 3$, so $(M_1, M_2) = (1, 3)$.

Proposition 5.3 (The abstract nested pair does not determine confinement)**.** *Present the target space first by*

$$10 \mapsto u, \qquad 01 \mapsto v, \qquad 11 \mapsto u + v,$$

and then by

$$11 \mapsto u, \qquad 10 \mapsto v, \qquad 01 \mapsto u + v.$$

The two inner codes have the same associated nested pair, the same unlabelled set of helper supports, and the same complete relative-weight hierarchy. Their inner-dual weight multisets are respectively $\{2, 3, 5\}$ *and* $\{3, 3, 4\}$*. Hence their additive outer-distance costs are respectively*

$$(M_1 + d(I^\perp), M_2 + d(I^\perp)) = (3, 5) \qquad \textit{and} \qquad (M_1 + d(I^\perp), M_2 + d(I^\perp)) = (4, 6).$$

Proof. Use the graph-code realization

$$I^{\perp} = \{(-\alpha^{-1}(x), x) : x \in D\},$$

where α is the displayed identification of the two-dimensional target space with D. Adding the target weight to the helper weight gives the two listed multisets directly. Thus $d(I^{\perp})$ is two in the first presentation and three in the second. Adding this distance to $(M_1, M_2) = (1, 3)$ gives the two threshold sequences by Theorem 4.8. □

These are limits on representation, not on executability. Keeping the functional labels avoids the second loss, and keeping minimizing lifts permits the exact recursion to return the original coefficient systems. The next section turns that closure law into an optimizer.

6 Exact recovery optimization

The closure law in Theorem 4.3 supplies an exact compiler state: its coordinates are the intermediate functionals that scalar summaries discard. ergodis (Exact Recovery, Global Optimization, and Invariant Synthesis) compiles and evaluates that state. Numerical tables store the fibrewise minima; separately stored predecessor labels and minimizing lifts reconstruct a coefficient-level witness for every optimum.

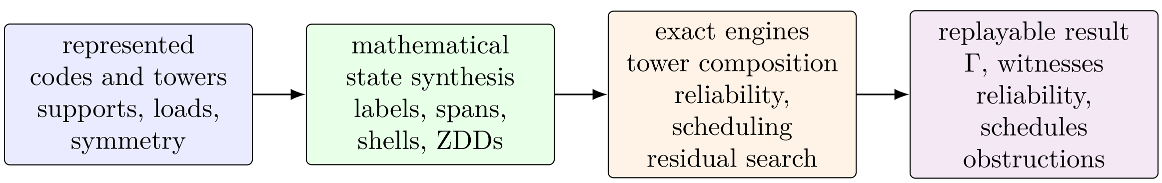

Figure 2: ergodis synthesizes the exact state exposed by the mathematics before search begins. Labelled coset costs drive tower composition, support families drive reliability, and graded load states drive scheduling. For attained fibres, stored lifts expand an optimum to a coefficient-level witness or obstruction; arbitrary remaining side constraints may be passed to CP-SAT. The mathematical results do not depend on executing this pipeline.

We first state the algorithm in flattened $\mathbb{F}_q$-linear coordinates. Let T have dimension t, let the outer message space have dimension m, and let the inner label space have dimension s. For the ith outer coordinate, write

$$A_i : \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^s) \longrightarrow \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$$

for its contribution to the outer restriction map, and let $\lambda_i(b)$ be the prescribed-coset support cost of the inner label b. Different inner blocks may have different cost functions.

Proposition 6.1 (Exact hierarchical recovery optimizer)**.** *For $c \in \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$, define*

$$D_0(c) = \begin{cases} 0, & c = 0, \\ +\infty, & c \neq 0, \end{cases} \qquad D_i(c) = \min_b \big(D_{i-1}(c - A_i b) + \lambda_i(b)\big). \tag{30}$$

Then $D_N(c)$ is the exact prescribed-coset support cost of the concatenated code at label c. Storing a minimizing label b and predecessor state at each finite entry returns an exact block-labelled recovery witness, not only its scalar cost. The same recursion applies to the target-normalized formula after adjoining its target-image labels to the state.

If every label is enumerated, put $Q = q^{mt}$ and $S = q^{st}$. The direct table algorithm uses $O(NQS)$ min–sum transitions, $O(Q)$ working cost memory, and $O(NQ)$ predecessor entries when all witnesses are retained. Unreachable labels, quotient coordinates, projectively equivalent columns, and repeated block types may be compiled out without changing the optimum or the replayed witness.

Proof. After i blocks, $D_i(c)$ is the least sum of inner support costs among labels $(b_1, \dots, b_i)$ satisfying $\sum_{j \le i} A_j b_j = c$. This invariant is true at $i = 0$. Conditioning on the last label b_i gives exactly (30), so induction proves the claim. The leaf supports lie in disjoint blocks, hence their costs add. Backtracking the stored minimizers reconstructs the labels; the inner prescribed-coset witnesses then reconstruct the coefficient-level recovery system. There are Q states and S candidate labels per block, which gives the stated bounds. The target-normalized case is the same invariant applied to the paired labels in (15). □

The resulting table supplies all exact helper costs needed by the finite confinement formula, including its zero and nonzero outer-functional sectors. It can therefore emit a replayable recovery witness when a threshold fails, or a complete exhausted-state certificate when the requested label is unreachable. Iterating the compiler through a tower preserves the intermediate labels required at the next level; collapsing each table to one minimum would destroy precisely this compositional information.

The same support functions also lead to an exact capacity-aware repair scheduler. Suppose demand i has at most A admissible recovery choices. Choice a consumes a nonnegative integer load vector $\ell_{i,a} \in \mathbb{Z}^H_{\ge 0}$ on H helpers, and capacities are $\boldsymbol{c} = (c_1, \dots, c_H)$. A demand may be skipped. This model distinguishes support occupancy from coefficient-weighted download and permits different capacities at different helpers.

Proposition 6.2 (Exact capacitated repair scheduling)**.** *Let $\mathcal{S} = \prod_{h=1}^{H} \{0, \dots, c_h\}$. A dynamic program indexed by the current load $u \in \mathcal{S}$, with transitions $u \mapsto u + \ell_{i,a}$ whenever the result is within capacity and with a skip transition, returns the maximum number of simultaneously served demands and an exact assignment witness. It uses*

$$O\left(DAH \prod_{h=1}^{H} (c_h + 1)\right) \quad \textit{time and} \quad O\left(\prod_{h=1}^{H} (c_h + 1)\right) \quad \textit{working memory,} \tag{31}$$

apart from stored predecessors.

If there are positive integer weights w_h and a common $M > 0$ such that $w \cdot \ell_{i,a} = M$ for every admissible choice, a state serving exactly r demands lies on the graded shell

$$\mathcal{S}_r = \{u \in \mathcal{S} : w \cdot u = rM\}.$$

Thus the full box may be replaced by the union of its reachable shells. Two distinct states on one shell are incomparable in the componentwise order, so ordinary Pareto-dominance pruning cannot remove either one. The grading is therefore both an exact compression and a certificate of the limit of that common pruning rule.

Proof. After processing the first i demands, associate to every reachable load the largest served count and a predecessor realizing it. The skip and assignment transitions enumerate exactly the feasible decisions for demand $i + 1$; induction gives optimality and witness recovery. Scanning at most A choices across H coordinates for every demand and box state gives (31). Under the grading hypothesis, additivity places every r-repair load on $\mathcal{S}_r$. If $u \le v$ componentwise and both lie on the same shell, positivity of w forces $u = v$, proving the antichain assertion. □

6.1 Measured solver boundary

The scientific role of the implementation is the exact executable semantics above. The measurements in this subsection assess engineering reach and the boundary with a general solver; they are not mathematical evidence or the principal contribution of the software.

The specialized dynamic programs are most useful when compilation exposes a small algebraic state space. They are not replacements for a general constraint solver when the residual model contains unrelated side constraints. The fair deployment boundary is therefore:

$$\text{algebraic compilation} \longrightarrow \begin{cases} \text{specialized exact DP}, & \text{when the compiled state is sufficient,} \\ \text{CP-SAT}, & \text{for the remaining general constraints.} \end{cases}$$

Direct CP-SAT is an appropriate end-to-end baseline. A second, *structured CP-SAT* baseline receives the same symmetry reduction and algebraic preprocessing as ergodis, isolating the value of the specialized state engine from the value of compilation itself.

The benchmark suite separates the algebraic compiler from the residual state engine. On four exact capacitated-scheduling profiles, ergodis was 2.5–377 times faster than direct CP-SAT and 2.5–373 times faster than structured CP-SAT receiving the same safe preprocessing. The represented tower benchmark shows where the composition theorem itself changes scale: at depth six and fanout four, ergodis expands an exact coefficient witness over 4096 leaves in 766 μs, while direct and labelled-table CP-SAT take 264 s and 6.19 s, respectively. The minute-scale direct entry is one completed deterministic solve; the shorter entries are interleaved medians.

The same compilation pattern has been tested through six bundled application examples. They cover recursive Ceph-style XOR repair, Azure-style locally repairable code (LRC) batching, precedence-constrained repair directed acyclic graphs (DAGs), quasi-cyclic low-density parity-check (QC-LDPC) search, vector/subpacketized repair, and MDS checkpoint recovery under node and rack capacities. Each formulation-specific control reproduces the same exact support family, optimum, or feasibility verdict, as applicable. The narrowest gap is 8 times, roughly one order of magnitude; the structure-dominated examples reach three to five orders of magnitude. Table 1 records the bounded instances. These comparisons do not include commercial solvers or claim a universal ranking.

application example	bounded instance	matched exact control	ergodis	control	ratio
Ceph XOR	8 diamonds, 256 supports	Graphillion ZDD closure	102 μs	864 μs	8×
Azure LRC	100,000 demands	HiGHS counted model	$< 1\,\mu$s	4877 μs	174,000×
repair DAG	3 × 21 tasks	CP-SAT interval model	2 μs	19,218 μs	7,881×
QC-LDPC	lift 50,000, weight 4	CryptoMiniSat native XOR	1517 μs	509,306 μs	336×
vector repair	64 nodes, 2 symbols/node	CryptoMiniSat native XOR	10 μs	48,691 μs	4,717×
GPU MDS	10,000 shards, 64 failures	OR-Tools max-flow	100 μs	103,061 μs	1,029×

Table 1: Exact comparisons for the bundled application examples. Times include construction of the application state or control model; every row agrees on the exact support family, optimum, or feasibility verdict being compared. Controls differ by application and are named explicitly; ratios use the unrounded timings.

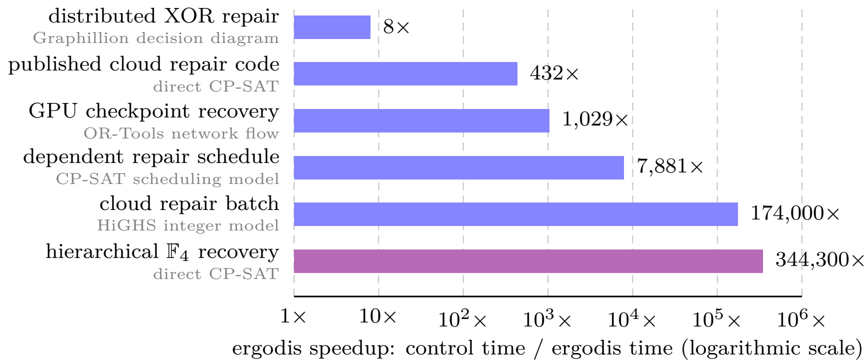

Figure 3: How many times faster ergodis is than matched exact controls for representative applications and the theorem-driven tower computation. Longer bars mean larger ergodis speedups. Each formulation-specific control is printed beneath its workload; full instances appear in Table 1, with the published cloud-code check below. The distinguished hierarchical $\mathbb{F}_4$ row compares labelled min–sum composition with direct CP-SAT and isolates the reduction supplied by the closure theorem. Controls differ by row, so this is not a general solver ranking.

As a code-native check, ergodis also analyzes Jin and Fu's published $\mathbb{F}_4$ Hamming-outer family [14, Corollary 5.4]. For its binary $[4095, 2718, 6; 2]$ LRC, all backends obtain the same exact confinement cost; ergodis takes 231 ms, versus 100 s for direct CP-SAT and 82 s for labelled-table CP-SAT. Exact inputs, outputs, tool versions, peak memory, replay commands, and raw timing samples are in `ergodis/evidence/benchmarks.json` and `ergodis/BENCHMARKS.md`; the two minute-scale CP-SAT entries are single completed deterministic solves.

The largest gains arise when the algebraic reductions expose a low-dimensional reachable set inside a much larger raw model. ergodis solves that state directly when it is sufficient and otherwise supplies an exact preprocessor for a residual CP-SAT model, retaining recovery witnesses and confinement thresholds in either case.

The optimizer changes no mathematical hypothesis: it evaluates the labelled state proved sufficient above. We now return to consequences of the collapsed RGHW threshold, beginning with ordinary generalized weights and extremal inner codes.

7 Consequences of the threshold

The fixed-target relative weights sit between ordinary generalized weights and worst-case cooperative locality. Let $d_e(C^\perp)$ denote the eth generalized Hamming weight of $C^\perp$. For an e-set $P \subseteq E$, let $\kappa_C(P)$ be the least $|H|$ for which some e-dimensional subcode $A \leq C^\perp$, supported on $P \cup H$, restricts isomorphically to $\mathbb{F}_q^P$. Set $\kappa_C(P) = \infty$ when no such subcode exists. Thus $\kappa_C(P)$ is the minimum number of helpers needed to recover all coordinates in P simultaneously.

Theorem 7.1 (Best-target generalized weight and first escape cost)**.** *For $1 \leq e \leq \dim C^\perp$,*

$$\min_{\substack{P \subseteq E \\ |P|=e}} \kappa_C(P) = d_e(C^\perp) - e. \tag{32}$$

Consequently, if every e-set is recoverable and $r_e^{\mathrm{coop}}(C) = \max_{|P|=e} \kappa_C(P)$, then

$$r_e^{\mathrm{coop}}(C) \geq d_e(C^\perp) - e.$$

For a fixed inner code in the eventual concatenation regime of Theorem 4.7, the least first nonconfinement cost over all e-coordinate target sets is

$$d_e(C^\perp) - e + d(C^\perp). \tag{33}$$

Proof. A normalized identity system on P spans an e-dimensional subcode of $C^\perp$ whose support is P together with its helper union. Hence $d_e(C^\perp) \le e + \kappa_C(P)$.

Conversely, let $A \le C^\perp$ have dimension e and support size $d_e(C^\perp)$. An information set P of a generator matrix of A has size e, and restriction $A \to \mathbb{F}_q^P$ is an isomorphism. Row reduction on these columns gives normalized identity equations whose helper union is $\operatorname{supp}(A) \setminus P$. This proves (32); the cooperative-locality inequality follows by taking the maximum instead of the minimum.

It remains to justify the escape formula when the target columns indexed by P are dependent. Put $U = \operatorname{im} G_P$. A normalized identity system on $\mathbb{F}_q^P$ and a normalized recovery system on U have the same minimum helper union. Indeed, restricting an identity system along a linear section $\alpha : U \to \mathbb{F}_q^P$ gives a system on U. Conversely, given a system on U, write

$$a = \alpha(G_P a) + \big(a - \alpha(G_P a)\big) \qquad (a \in \mathbb{F}_q^P).$$

The second summand lies in $\ker G_P$ and hence is a target-only dual equation, so adjoining these kernel equations recovers the full identity without adding helpers. Thus the common minimum is $\kappa_C(P)$.

Now repeat the block-functional argument of Theorem 4.7. Once the outer-functional case is excluded, the target block of any identity system uses at least $\kappa_C(P)$ helpers. If the system is nonconfined, a nonzero equation map in another block has union support at least $d(C^\perp)$; the two supports are disjoint. Conversely, attach a minimum-weight word of $C^\perp$ in a second block through any nonzero linear functional on $\mathbb{F}_q^P$. Hence the first nonconfinement cost for this P is $\kappa_C(P) + d(C^\perp)$. Minimizing over P and using (32) proves (33). □

Before bounding the MDS thresholds, we record what the coefficient layer adds even when the support clutter is completely uniform.

Theorem 7.2 (Minimum-weight MDS recovery equations span the dual)**.** *Let $C \le \mathbb{F}_q^E$ be an $[n,k]_q$ MDS code with $1 \le k < n$, and let $x \in E$. Then*

$$\operatorname{span} \mathcal{P}_x^{(\le k)}(C) = C^\perp,$$

and their supports form the complete k-uniform clutter on $E \setminus \{x\}$.

Proof. The dual has dimension $n-k$ and distance $k+1$. For each $(k+1)$-set T, restriction $C^\perp \to \mathbb{F}_q^{E\setminus T}$ has a nonzero kernel, whose words have support exactly T by the distance bound. Thus all stated supports occur. Fix a $(k-1)$-set $Q \subseteq E \setminus \{x\}$ and take the normalized word on $\{x,t\} \cup Q$ for each $t \in E \setminus (\{x\} \cup Q)$. Their private coordinates t make these $n-k$ words independent, hence a basis of $C^\perp$. □

The support clutter depends only on the MDS parameters, whereas the normalized equations retain the represented dual code.

Put $k = \dim I$ and $b = \operatorname{rank} G_J$. The exact sequence and the relative and ordinary Singleton bounds give

$$M_t(D_P, K_P) \le b - \ell + t, \qquad d(I^\perp) \le k + 1. \tag{34}$$

Theorem 7.3 (MDS ceiling and rank-one rigidity)**.** *For* $1 \le t \le \ell$,

$$M_t(D_P, K_P) + d(I^{\perp}) \le 2k - \ell + t + 1. \tag{35}$$

Suppose I is MDS, and put

$$u = \min\{k, |P|\}, \qquad b = \min\{k, |J|\}.$$

Then $\ell = u + b - k$. If $\ell \ge 1$, then, for $1 \le t \le \ell$,

$$M_t(D_P, K_P) = k - u + t, \qquad M_t(D_P, K_P) + d(I^{\perp}) = 2k - u + t + 1. \tag{36}$$

If $|J| \ge k$, then $\ell = u$ and these values attain the ceiling in (35) at every rank. Conversely, when $\ell \ge 1$, equality in (35) at $t = 1$ forces I to be MDS and forces equality at every rank.

Proof. The ceiling follows from (34) and $b \le k$. For an MDS code the column matroid is uniform, so $\operatorname{rank} G_P = u$, $\operatorname{rank} G_J = b$, $\ell = u + b - k$, and, for $H \subseteq J$ with $|H| = h \le k$,

$$\dim\bigl(\operatorname{span} G_P \cap \operatorname{span} G_H\bigr) = \max\{0, u + h - k\}.$$

Thus Proposition 3.4 gives $M_t = k - u + t$ for $1 \le t \le \ell$; here $k - u + t \le b \le |J|$. Now $d(I^{\perp}) = k+1$, and if $|J| \ge k$, then $b = k$ and $\ell = u$, proving the remaining direct assertions.

Equality at $t = 1$ forces equality in both Singleton bounds and $b \le k$. Hence $d(I^{\perp}) = k + 1$, so I is MDS, and $M_1 = b - \ell + 1$. Strict growth then gives $M_t \ge b - \ell + t$, while relative Singleton gives the reverse inequality; every rank is extremal. □

Thus extremality is detected at rank one: equality in the first threshold forces the inner code to be MDS and forces every later rank to attain its ceiling. No separate higher-rank hypothesis is needed.

We next record the global parameters and the density of the transferred data. Let the inner code have parameters $[m, k, d]_q$, so $[L : \mathbb{F}_q] = k$. Let $K_N = \dim_L O_N$ and $D_N = d(O_N)$.

Corollary 7.4 (Positive-density realization)**.** *Assume $d(O_N^{\perp}) \to \infty$. Below any fixed threshold supplied by Theorem 4.7 or Theorem 4.8, for all sufficiently large N every inner normalized recovery system and exact helper support is copied in each of the N blocks. Each fixed inner coordinate type has density $1/m$ among the concatenated coordinates; the union of the types in P has density p/m.*

In particular, if $r < M_1(D_P, K_P) + d(I^{\perp})$, then all cost-$\le r$ systems for all nonzero subspaces of W_P are copied simultaneously in every block. Hence the complete bounded local recovery data through radius r occur on the stated positive-density coordinate classes.

The concatenated code has length mN, dimension kK_N, and minimum distance at least dD_N. If

$$\frac{K_N}{N} \longrightarrow R > 0, \qquad \liminf_N \frac{D_N}{N} \ge \delta > 0,$$

then the concatenated family has rate kR/m and relative distance at least $d\delta/m$. Outer families satisfying all three hypotheses exist over every finite field L.

Proof. The transfer statement is the zero-extension bijection in the confinement theorems, applied independently in every block. There are mN coordinates, and each inner coordinate occurs once in each block, which gives the density claims. The blockwise inner encoder is injective, so restriction of scalars gives dimension kK_N. A nonzero outer word has at least D_N nonzero symbols, each encoding to an inner word of weight at least d. This proves the distance and asymptotic assertions.

For the existence statement, write $Q = |L|$. Choose $0 < R < 1$ and positive $\delta, \delta^{\perp}$ such that

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^{\perp}) < R,$$

where H_Q is the Q-ary entropy function. The first-moment argument for a random K_N-dimensional L-linear code bounds the expected numbers of nonzero primal and dual words below the two cutoffs, up to subexponential factors, by

$$Q^{N(H_Q(\delta)+R-1)} \qquad \text{and} \qquad Q^{N(H_Q(\delta^{\perp})-R)},$$

respectively. Both tend to zero by the displayed inequalities, so one may choose a sequence with rate tending to R, primal relative distance at least δ, and dual relative distance at least $\delta^{\perp}$; see also [20, Chapter 17]. $\square$

Exact support transfer also preserves the fractional load-balancing region used in service-rate problems. We state the consequence because it requires only the inclusion-minimal recovery sets, whereas the theorem transfers all normalized coefficients as well.

Fix one inner target block with helper set J. Let $\mathcal{A}$ be a finite set of demands on that block, with demand a represented by a nonzero target subspace $T_a \leq W_P$, and let $\mathcal{M}_a^{(\leq r)}$ be the inclusion-minimal exact helper supports of size at most r for $a \in \mathcal{A}$. For a nonnegative helper-capacity vector c, define

$$\Lambda_{\mathcal{A}}^{(\leq r)}(c) = \left\{ \lambda \geq 0 : \begin{array}{l} \text{there are } f_{a,R} \geq 0 \text{ with } \lambda_a = \sum_{R \in \mathcal{M}_a^{(\leq r)}} f_{a,R}, \\ \displaystyle\sum_a \sum_{R \ni j} f_{a,R} \leq c_j \text{ for every helper } j \end{array} \right\}.$$

This is the bounded service-rate region of the recovery-set system [22, 7].

Corollary 7.5 (Transfer of bounded service-rate regions)**.** *Let $O \leq L^N$ be an outer code with $N \geq 2$ whose projection onto the target block is nonzero. Suppose*

$$r < \Gamma_{j,T_a}(O, I) \qquad (a \in \mathcal{A}).$$

Transport capacities from the inner helper set to its copy in that block of $O \circ I$. Then the inner and concatenated bounded service-rate regions are equal. Capacities assigned outside the target block do not change this equality. In particular, the hypothesis follows from $d(O^{\perp}) > r + 1$ and the corresponding inner inequalities. Consequently, for an outer family with $d(O_N^{\perp}) \to \infty$, the equality holds for all sufficiently large N whenever those inner inequalities hold.

Proof. Theorem 4.1 confines every cost-$\leq r$ system for every demand. Thus restriction and zero-extension copy every inclusion-minimal support and create no cross-block minimal support. The helper bijection therefore transports the feasible variables $f_{a,R}$ and every capacity inequality in both directions. Allowing the full upward-closed recovery-set family gives the same region: flow assigned to a nonminimal set can be moved to a contained minimal set and weakly decreases every helper load. Thus cross-block supersets and capacities outside the target block are irrelevant. $\square$

The next family realizes all three information levels in closed form: its RGHWs give minimum recovery costs, its ambient dual distance gives the gated confinement thresholds, and its projective rank distribution gives reliability beyond the RGHW hierarchy.

8 The projective-simplex family

This non-MDS family realizes the paper's information hierarchy in one model. Its relative weights determine the minimum costs, its ambient inner dual gives the collapsed confinement thresholds, and the full projective rank distribution determines reliability. Thus the family shows both what the RGHW hierarchy captures and what it forgets. Assume $m \geq 2$, put

$$N = \frac{q^m - 1}{q - 1},$$

and let A be an $m \times N$ matrix containing one representative of every point of $\mathrm{PG}(m-1, q)$. Define the inner code by

$$I_m^{\perp} = \{(a, aA) : a \in \mathbb{F}_q^m\}. \tag{37}$$

Take the first m coordinates as targets and the projective coordinates as helpers. The associated nested pair is isomorphic to

$$0 \subseteq S_m = \{aA : a \in \mathbb{F}_q^m\},$$

where S_m is the q-ary simplex code. The generalized-weight formula for S_m is classical; in fact every t-dimensional simplex subcode has the same support size [26]. We include the short projective count to connect that hierarchy to the confinement and reliability statements below.

Theorem 8.1 (Projective-simplex thresholds)**.** *For $1 \leq t \leq m$,*

$$M_t(S_m, 0) = \frac{q^m - q^{m-t}}{q - 1}, \tag{38}$$

and

$$d(I_m^{\perp}) = q^{m-1} + 1. \tag{39}$$

Hence

$$\Gamma_t := M_t(S_m, 0) + d(I_m^{\perp}) = \frac{q^m - q^{m-t}}{q - 1} + q^{m-1} + 1. \tag{40}$$

In outer families with dual distance tending to infinity, Γ_t is the eventual first nonconfined helper cost at recovered dimension t. The family is non-MDS for $m \geq 2$; for $m \geq 3$ the successive increments of Γ_t are unequal.

Proof. The target restriction of a word $(a, aA) \in I_m^{\perp}$ is a, and no nonzero dual word has zero target restriction. Hence normalized recovery systems correspond, with support preserved, to subcodes of S_m; this proves the asserted identification of the associated pair.

Let $U \leq \mathbb{F}_q^m$ have dimension t. The common zero coordinates of the subcode UA are the projective points in $U^{\perp}$, a subspace of projective dimension $m - t - 1$. Thus

$$|\operatorname{supp}(UA)| = N - \frac{q^{m-t} - 1}{q - 1} = \frac{q^m - q^{m-t}}{q - 1},$$

independently of U. This proves (38). Every nonzero simplex word aA has weight q^{m-1}, while a nonzero target vector a has weight at least one. Equality is attained by a coordinate vector, which proves (39). The confinement formula is Theorem 4.8. The non-MDS and increment assertions follow from the displayed parameters. $\square$

question	retained information	exact answer
minimum rank-t helper union	RGHW hierarchy	$\dfrac{q^m - q^{m-t}}{q-1}$
eventual first nonconfined cost	RGHW plus $d(I_m^\perp)$	$\dfrac{q^m - q^{m-t}}{q-1} + q^{m-1} + 1$
rank-t recovery reliability	projective rank distribution	$R_t(s)$ in (42)

Table 2: The projective-simplex family separates the three questions. The first two use minimum-cost data; the stochastic law requires the distribution of ranks of failed projective sets.

This family also gives an exact stochastic law at every recovered dimension. Let each helper survive independently with probability s, and let F be the set of failed projective points. The available recovery equations are indexed by

$$\{a \in \mathbb{F}_q^m : aA|_F = 0\} = \big(\operatorname{span} F\big)^\perp,$$

which has dimension $m - \operatorname{rank}(F)$.

Theorem 8.2 (Projective rank reliability)**.** *The probability of retaining at least t independent target recovery equations is*

$$R_t(s) = \sum_{\substack{F \subseteq \mathrm{PG}(m-1,q) \\ \operatorname{rank}(F) \le m-t}} (1-s)^{|F|} s^{N-|F|}. \tag{41}$$

Writing $N_v = (q^v - 1)/(q-1)$, this equals

$$R_t(s) = \sum_{u=0}^{m-t} \begin{bmatrix} m \\ u \end{bmatrix}_q \sum_{v=0}^{u} \begin{bmatrix} u \\ v \end{bmatrix}_q (-1)^{u-v} q^{\binom{u-v}{2}} s^{N-N_v}. \tag{42}$$

Its endpoint terms are

$$R_t(s) = \begin{bmatrix} m \\ t \end{bmatrix}_q s^{M_t} + O(s^{M_t+1}) \qquad (s \to 0), \tag{43}$$

$$1 - R_t(s) = B_{m,m-t+1}(1-s)^{m-t+1} + O((1-s)^{m-t+2}) \qquad (s \to 1), \tag{44}$$

where

$$B_{m,r} = \begin{bmatrix} m \\ r \end{bmatrix}_q \frac{|\operatorname{GL}(r,q)|}{r!(q-1)^r}$$

is the number of unordered projective r-frames.

Proof. Recovery of t independent target combinations is possible exactly when $m - \operatorname{rank}(F) \ge t$, which gives (41). For a vector subspace $U \le \mathbb{F}_q^m$, let $\mathbb{P}(U)$ denote its projective points. Möbius inversion in the subspace lattice gives the total probability of failed sets spanning a fixed u-space U as

$$\sum_{V \le U} \mu(V,U) s^{N-N_{\dim V}}, \qquad \mu(V,U) = (-1)^{u-v} q^{\binom{u-v}{2}}, \qquad v = \dim V.$$

Here the total weight of all failed sets contained in $\mathbb{P}(V)$ is

$$\sum_{F \subseteq \mathbb{P}(V)} (1-s)^{|F|} s^{N-|F|} = s^{N-N_{\dim V}},$$

which accounts for the power of s before inversion. There are $\begin{bmatrix} m \\ u \end{bmatrix}_q$ choices for U and $\begin{bmatrix} u \\ v \end{bmatrix}_q$ subspaces $V \le U$ of dimension v. Summing over $u \le m - t$ proves (42).

For $s \to 0$, a smallest surviving set that permits rank-t recovery is the complement of the points in an $(m-t)$-dimensional vector subspace. It has size M_t, and there are $\begin{bmatrix} m \\ m-t \end{bmatrix}_q = \begin{bmatrix} m \\ t \end{bmatrix}_q$ such subspaces. For $s \to 1$, failure first occurs when the failed points span dimension $m - t + 1$. A smallest such set is a projective frame of that size. Counting its span and then its unordered projective bases gives $B_{m,m-t+1}$. ☐

At the same helper length, the simplex presentation is characterized by its first recovery cost. This is the length-N, multiplicity-one specialization of Bonisoli's classification of linear equidistant codes [4]; the proof below records the precise zero-column and projective-multiplicity conventions used here.

Proposition 8.3 (Equality in the first projective bound)**.** *Let B be any full-row-rank $m \times N$ matrix over $\mathbb{F}_q$, with $N = (q^m - 1)/(q-1)$. Then*

$$\min_{0 \ne a \in \mathbb{F}_q^m} \operatorname{wt}(aB) \le q^{m-1}.$$

Equality holds if and only if, up to column scaling and permutation, the columns of B represent every point of $\mathrm{PG}(m-1, q)$ *exactly once.*

Proof. For each nonzero column, the fraction of nonzero $a \in \mathbb{F}_q^m$ for which its entry in aB is nonzero is

$$\frac{q^{m-1}(q-1)}{q^m - 1}.$$

Averaging $\operatorname{wt}(aB)$ over $a \ne 0$ gives at most q^{m-1}, strictly less if B has a zero column. If the minimum attains this average, every nonzero linear combination has weight q^{m-1} and every projective hyperplane contains total column multiplicity $(q^{m-1} - 1)/(q-1)$.

Let c be the vector of projective column multiplicities. The point–hyperplane incidence matrix is square, and its real Gram matrix has the form $(r - \lambda)I + \lambda J$ with $r > \lambda$. It is nonsingular, so the constant hyperplane-sum equations have the unique solution $c = \mathbf{1}$. Conversely, the simplex matrix has constant nonzero weight q^{m-1}. ☐

9 Verification boundary

The paper has a paper-owned Lean 4 companion built against a pinned Mathlib revision. It formalizes the construction of the associated nested code pair from abstract target and helper generator maps. In the notation of (1), the checked statements are

$$K_P \le D_P, \qquad G_J(D_P) = W_P, \qquad G_J|_{D_P} : D_P \twoheadrightarrow W_P, \qquad \ker(G_J|_{D_P}) = K_P.$$

The corresponding declarations are

Paper statement	Lean declaration
$K_P \le D_P$	`helper_ker_le_helperCodeForTargetSpace`
$G_J(D_P) = W_P$	`map_helperCodeForTargetSpace_eq_recoverableTargetMessageSpace`
surjectivity	`recoverableTargetMap_surjective`
restricted kernel	`mem_ker_recoverableTargetMap_iff`

The companion uses Lean 4.32.0-rc1 and the Mathlib commit

`571b8a8e54219b4d393f75f4b8653fac08197fcc`.

The axiom audit of all four reviewer-facing declarations reports only

$$\{\texttt{Classical.choice}, \texttt{Quot.sound}, \texttt{propext}\}.$$

No coding-theory result is introduced as a Lean axiom.

The formal coverage stops at the exact sequence. In particular, Proposition 3.2, Theorems 3.3, 4.1, 4.7, 4.8, and 4.9, as well as Proposition 4.3, the strictness construction in Proposition 4.10 and the later consequences and examples, are human-proved and are not established by the paper-local Lean companion. The claim manifest records them as absent rather than treating the exact-sequence formalization as evidence for them.

The reliability polynomials in Proposition 5.1 follow from the displayed union-size table by inclusion–exclusion. No exhaustive computation is a premise of that proposition or of any theorem in the main proof chain. Machine-readable examples and replay checks may therefore be used to audit arithmetic without changing the logical dependence of the paper.

10 Conclusion

The central question is compositional: which recovery information must be retained before an inner code is placed inside an outer one? The canonical shortening–puncturing pair answers the single-block minimum-cost question. Its relative generalized Hamming weights are exactly the minimum helper unions for recovered subspaces of each dimension. They are not, however, sufficient recursive state for finite concatenation.

Exact finite transfer is captured by the prescribed-coset support functions indexed by outer functional labels. Minimizing their blockwise costs over the complete outer functional dual gives $\Gamma_{j,T}(O, I)$, the minimum cost of a nonconfined normalized recovery system. The same functions compose associatively by min–sum substitution through finite towers. The numerical tables carry fibrewise minima; retaining minimizing lifts alongside them transports optimizing coefficient-level witnesses. Rank-one probes characterize the numerical contextual quotient, and the rank-one bottleneck controls simultaneous confinement at every recoverable rank. Below that bottleneck, restriction and zero-extension preserve the complete bounded coefficient and support families, hence every reliability or scheduling law determined by them.

The additive threshold

$$M_t(D_P, K_P) + d(I^{\perp})$$

is the zero-functional sector of the exact formula. Outer dual distance makes it exact by excluding every nonzero-functional sector below the chosen radius. Restriction and zero-extension then preserve normalized equations and exact helper supports, not only a locality value. This coefficient-level bijection yields the positive-density and bounded service-rate transfers; the one-coordinate weighted formula can remain exact even when the ordinary outer-distance gate fails.

The separations identify the cost of each projection. The complete RGHW hierarchy determines minimum helper unions but not the overlap law governing bounded reliability. Even the binary image code, its dual, the abstract nested pair, the complete relative-weight hierarchy, and the unlabelled helper supports can agree while one fixed outer code distinguishes two extension-field presentations through their functional labels. A separate ambient-presentation example shows that $d(I^{\perp})$ supplies another degree of freedom invisible to the abstract pair. The projective-simplex family exhibits all three levels explicitly: RGHWs give its minimum costs, ambient dual distance gives its collapsed thresholds, and the projective rank distribution gives its reliability polynomial.

ergodis evaluates the closure theorem rather than serving as evidence for it. Its min–sum compiler stores labelled minima and predecessor lifts, returning exact hierarchical costs and replayable witnesses; its scheduler then optimizes the resulting recovery choices under heterogeneous capacities. The recorded matched exact-control comparisons measure this implementation boundary on declared instances and do not enter any proof.

Two extensions require different state. Disjoint simultaneous service is an integral packing problem rather than a minimum-union or fractional service-rate problem. Bandwidth-aware array-code repair lets each helper transmit a subspace of functionals, so Hamming support is no longer the relevant cost. Neither extension follows by relabelling the present invariant; each requires a new compositional cost model.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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